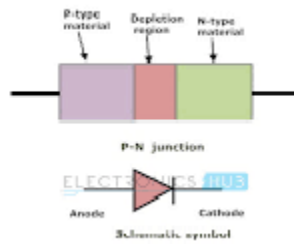
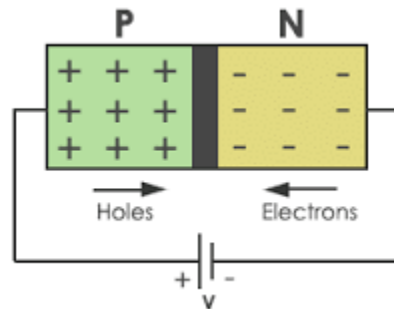


UNIT- I

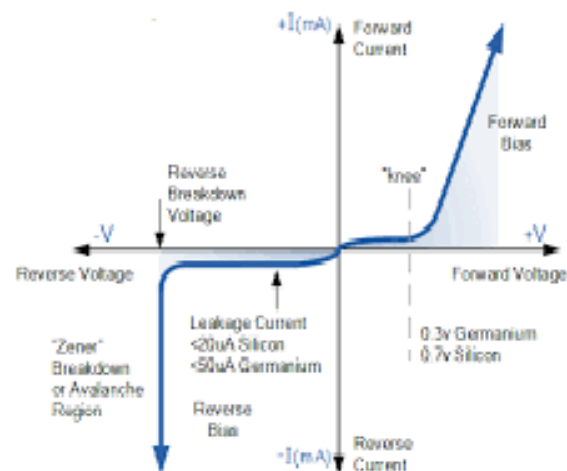
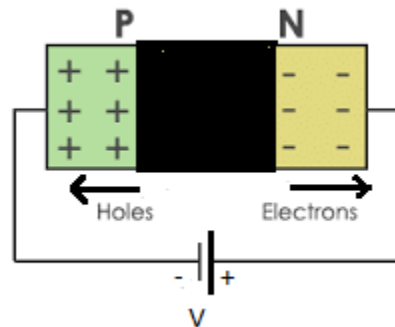
1. PN JUNCTION DIODE:



Forward biasing:

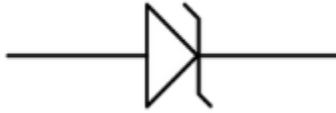


Reverse biasing:

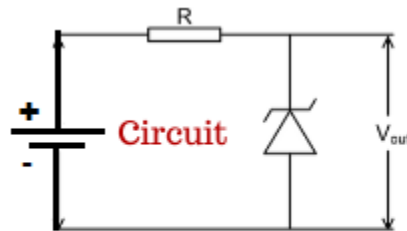


2. ZENER DIODE:

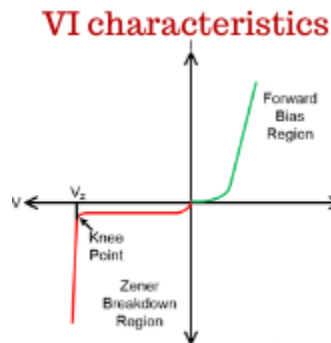
Symbol



Circuit Diagram:

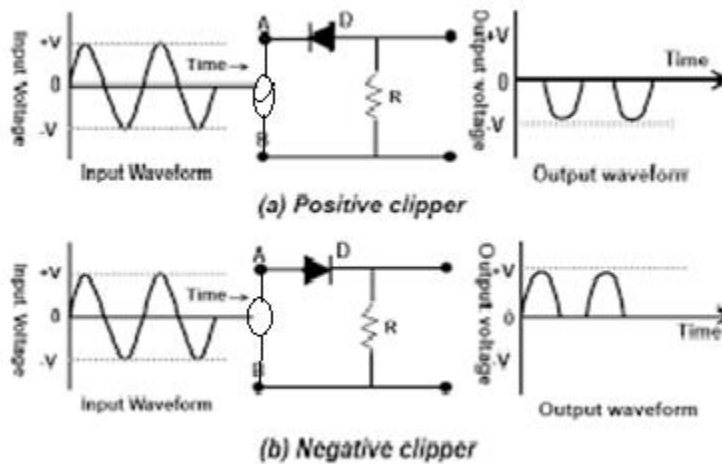


Graph:

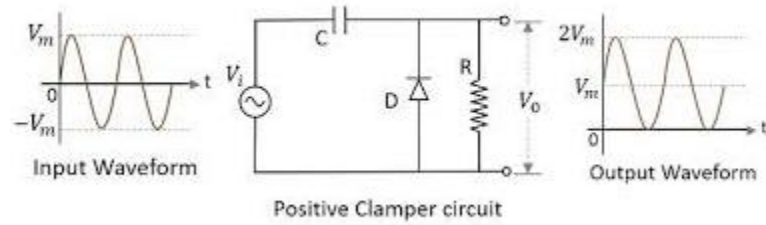


CLIPPER AND CLAMPER:

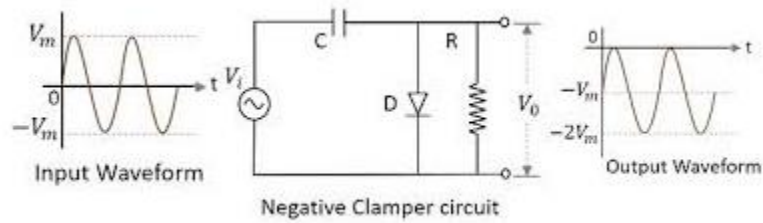
Clipper Circuit Diagram



Clamper Circuit Diagram



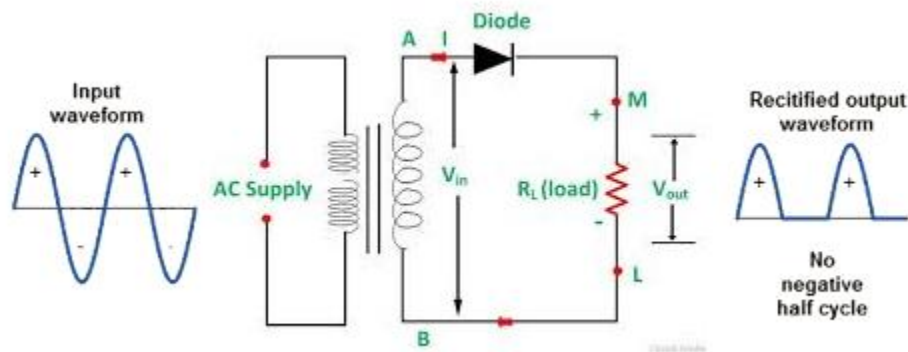
(a) Positive clamper



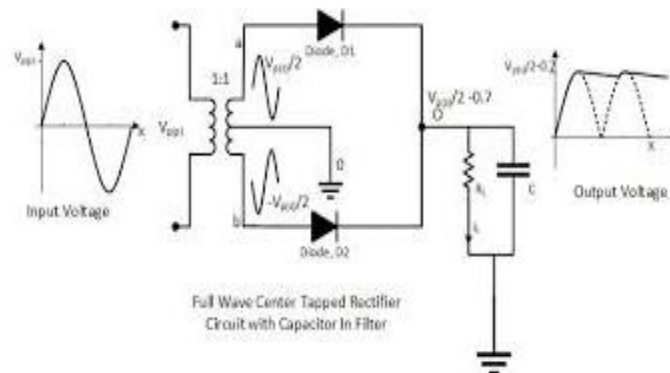
(b) Negative Clamper

3. RECTIFIER:

Halfwave rectifier:

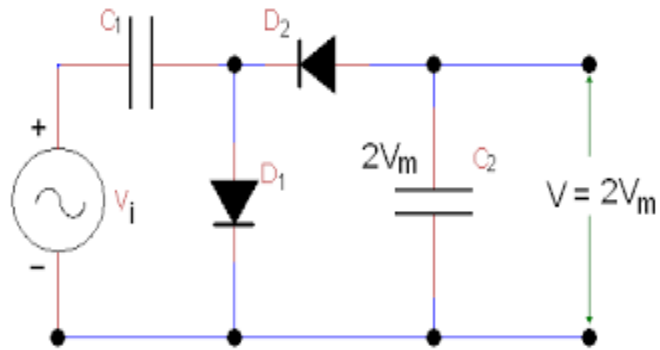


Fullwave Rectifier With Filter:

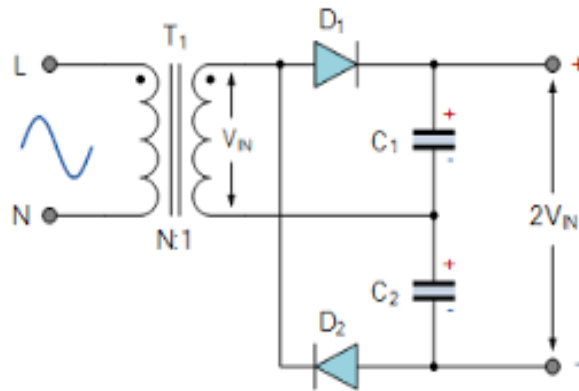


4. VOLTAGE MULTIPLER:

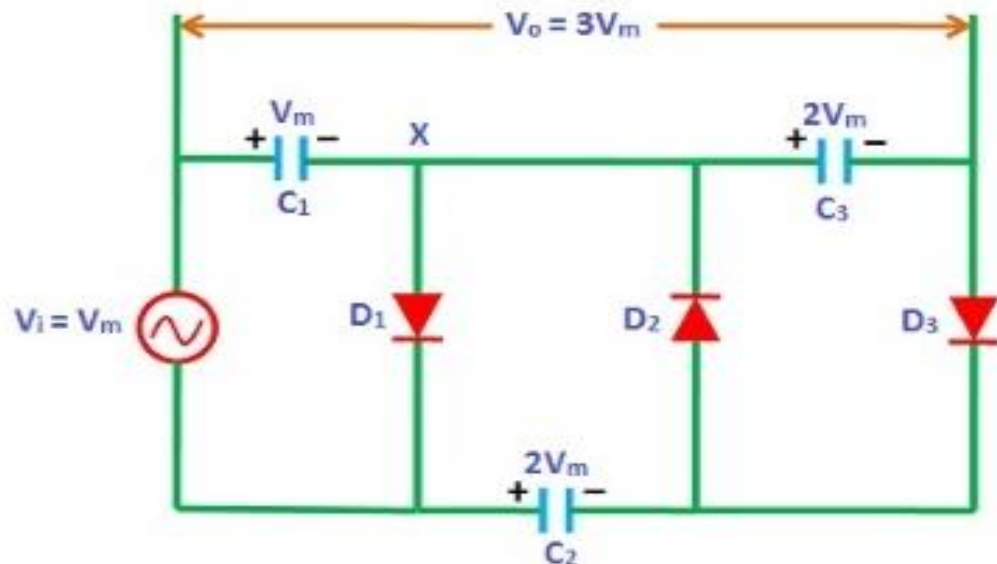
Half wave voltage doubler:



Full wave voltage doubler:

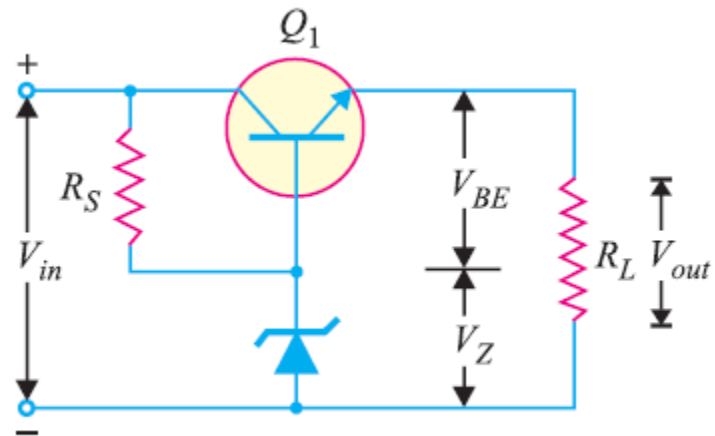


Voltage Tripler:

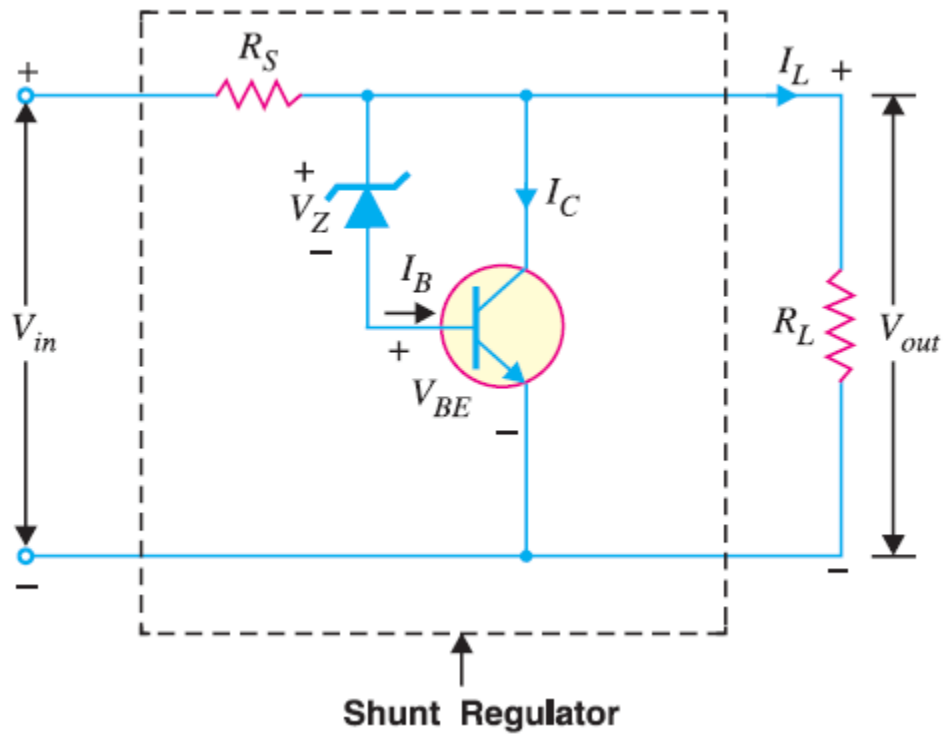


5. VOLTAGE REGULATOR:

Series Voltage Regulator:



Shunt Voltage Regulator:



6. OR and AND GATE USING DIODE:

OR Circuit:

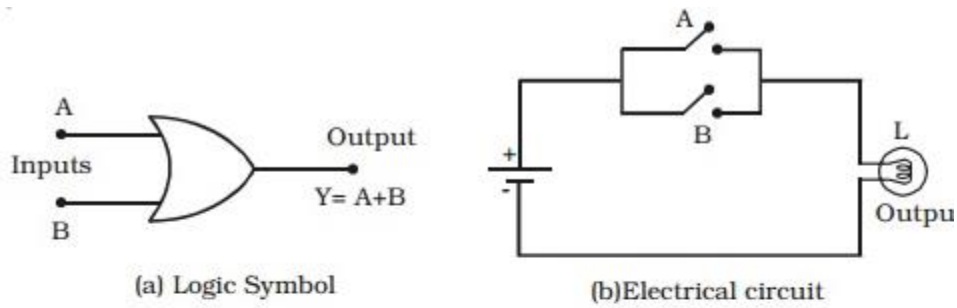


Fig OR gate

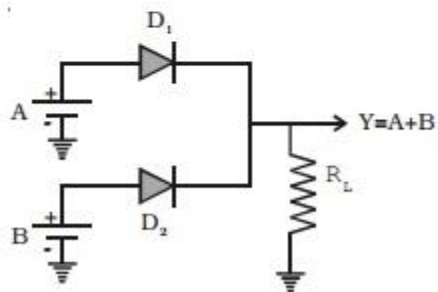


Fig OR gate using diodes

Table Truth table of OR gate

Inputs		Output
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

AND Circuit:

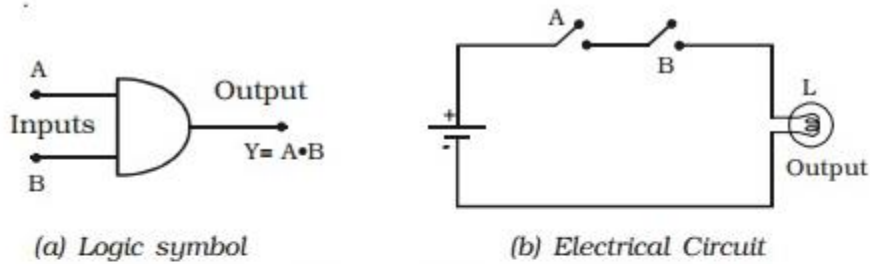


Fig AND gate

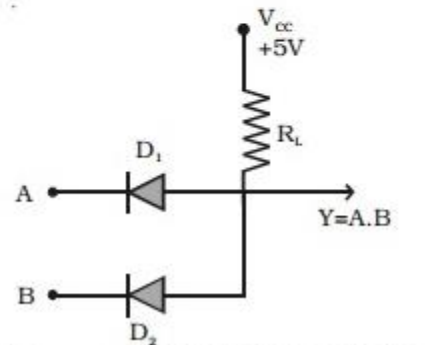


Fig AND gate using diodes

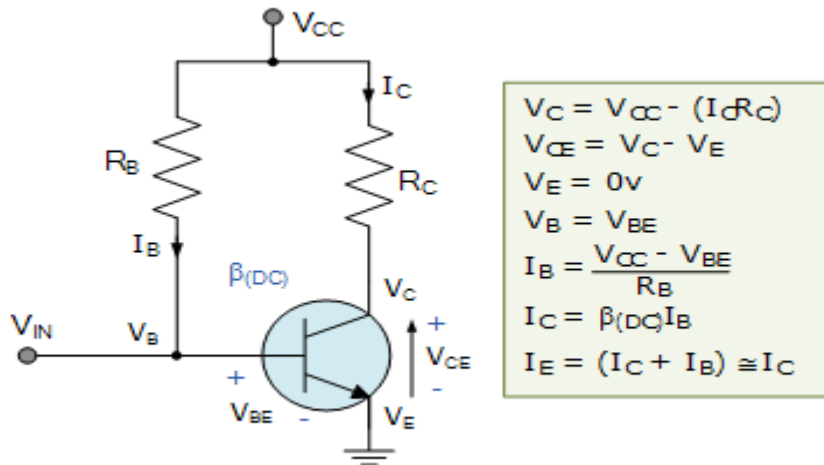
Table Truth table of AND gate

Inputs		Output
A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

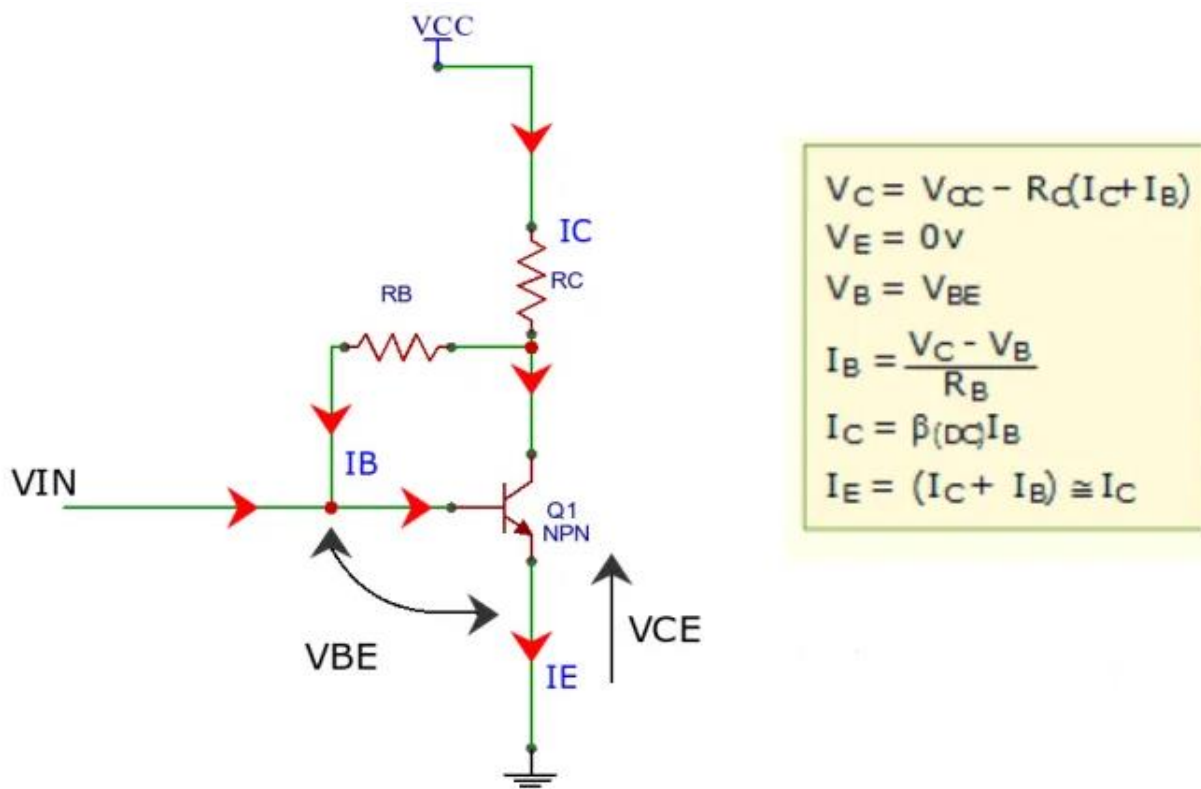
UNIT- II

1. TRANSISTOR BIASING:

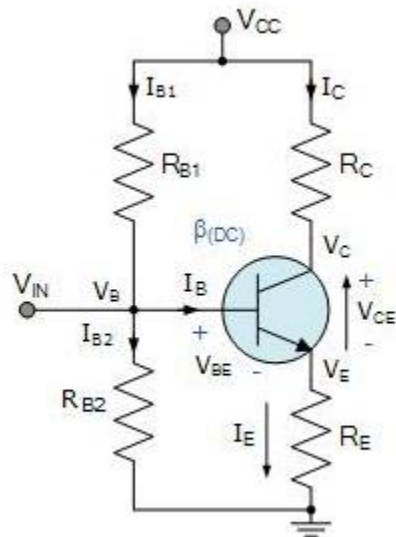
(a) Fixed bias:



(b) Collector to base bias:



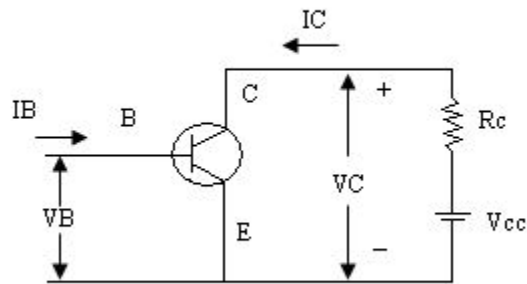
(c) Self or Voltage divider bias:



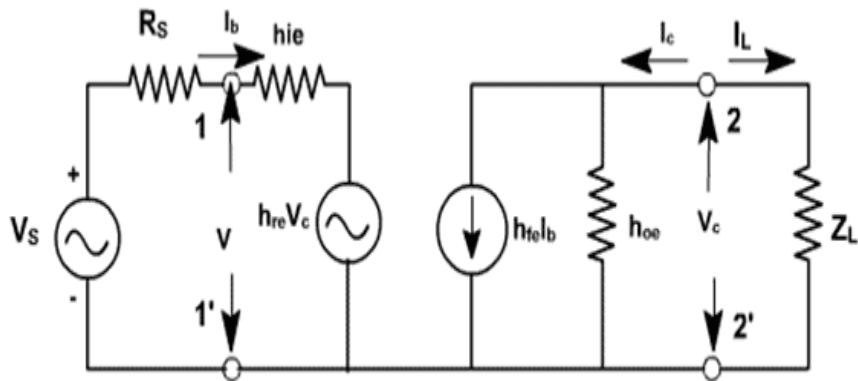
$$\begin{aligned}
 V_C &= V_{CC} - R_C I_C = V_E + V_{CE} \\
 V_E &= I_E R_E = V_B - V_{BE} \\
 V_{CE} &= V_C - V_E = V_{CC} - I_C (R_C + R_E) \\
 V_B &= V_{BE} + V_E = V_{RB2} = \left(\frac{R_{B2}}{R_{B1} + R_{B2}} \right) V_{CC} \\
 I_{B2} &= \frac{V_B}{R_{B2}} \\
 I_{B1} &= I_B + I_{B2} = \frac{V_C - V_B}{R_{B1}} \\
 I_B &= \frac{V_B}{R_B + (1 + \beta) R_E} \\
 I_C &= \beta_{(DC)} I_B \\
 I_E &= I_C + I_B = \frac{V_E}{R_E}
 \end{aligned}$$

2. H- PARAMETER COMMON EMITTER CONFIGURATION:

Equivalent circuit:



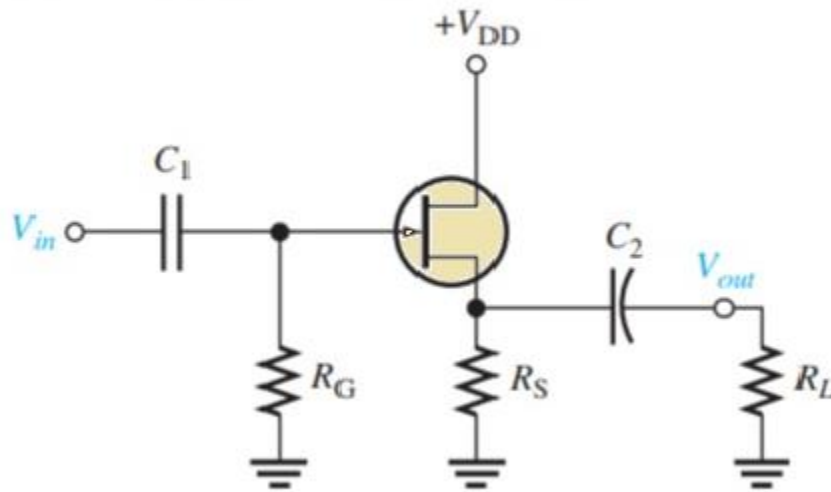
H – parameter:



3. JFET:

(a) Common Drain:

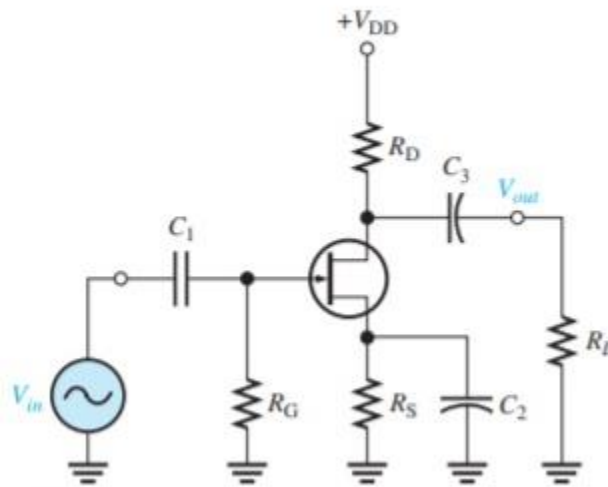
Common-Drain FET Amplifiers



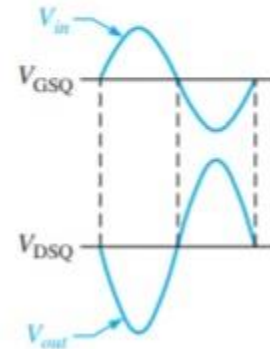
JFET common-drain amplifier (source-follower)

(b) Common Source:

JFET common-source amplifier

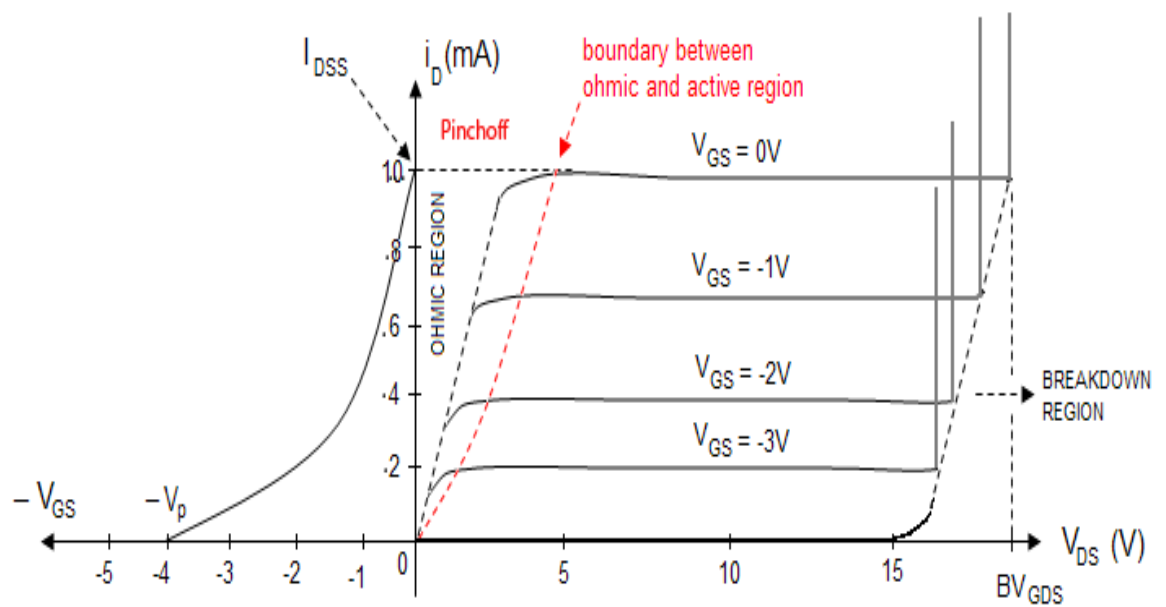
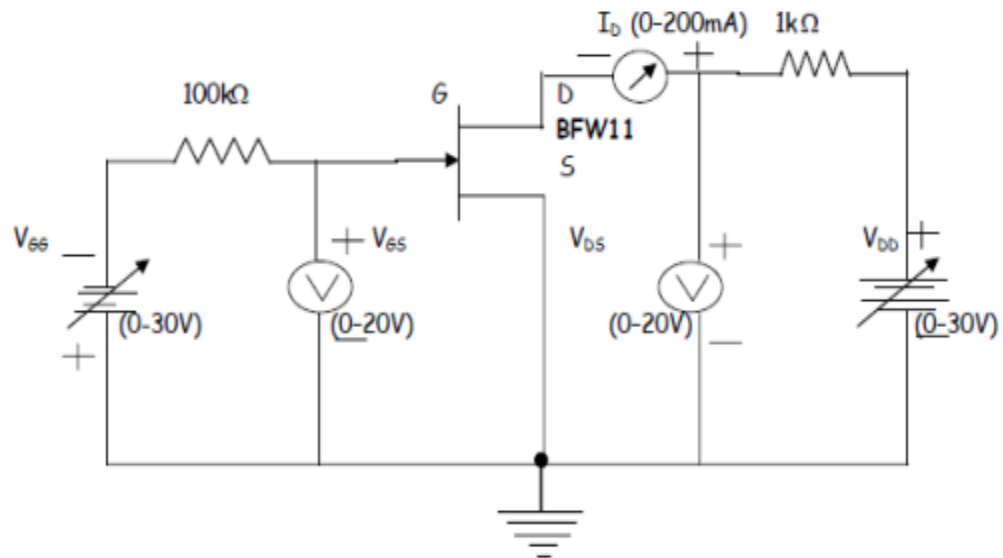


(a) Schematic



(b) Voltage waveform relationship

4. JUNCTION FIELD EFFECT TRANSISTOR:



Transfer characteristics

Drain characteristics